

Control Systems (EEE/ECE/INSTR F242)

Tutorial 2 (20 January 2026)

1. It is desired to balance a rolling ball of mass ' m ' on a tilting beam as shown in Fig. Q1. Assume that the motor input current i controls the torque linearly ($T_m = K_m i$) with negligible friction and the beam may be balanced near the horizontal ($\phi = 0$); with a small deviation in ϕ . Find the transfer function $X(s)/I(s)$. Given, J is the combined inertia of beam & shaft and ball has negligible inertia. Assume that forces acting on the ball are due to gravity and friction, and ' b ' is the coefficient of friction between ball and beam.

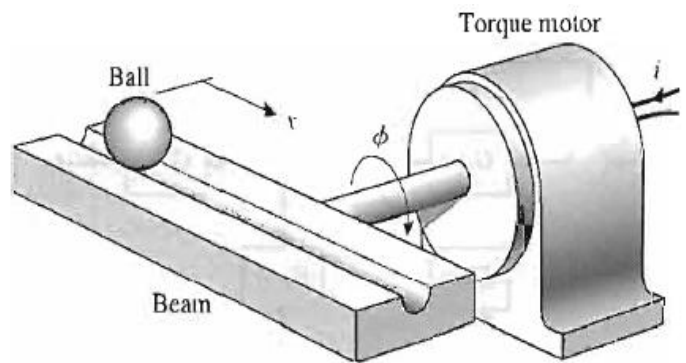


Fig. Q1

2. Find the transfer function $\theta_1(s)/T_1(s)$ for the system shown in Fig. Q2.

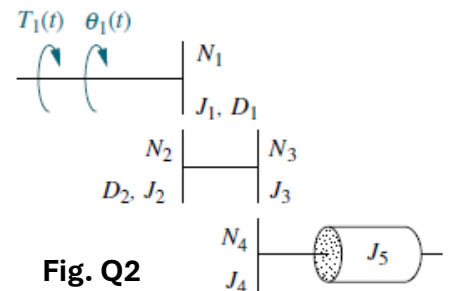


Fig. Q2

3. For the block diagram shown in Fig. Q3(a) and Q3(b), find closed loop transfer function $G(s) = C(s)/R(s)$, using block diagram reduction technique.

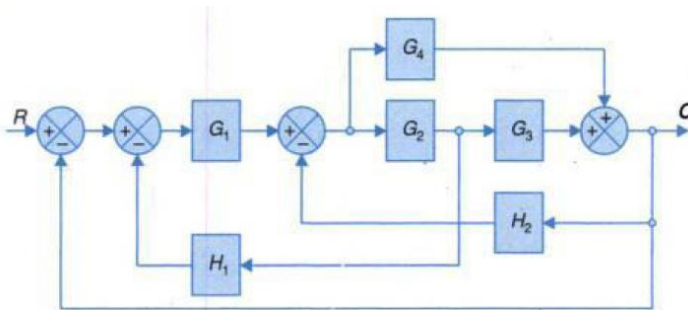


Fig. Q3(a)

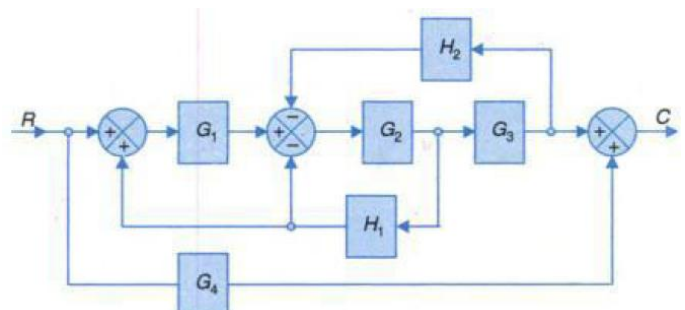


Fig. Q3(b)

4. Write the differential equation of motion describing the mechanical system shown in Fig. Q4. Also, draw the analogous electrical model using force-voltage and force-current analogy.

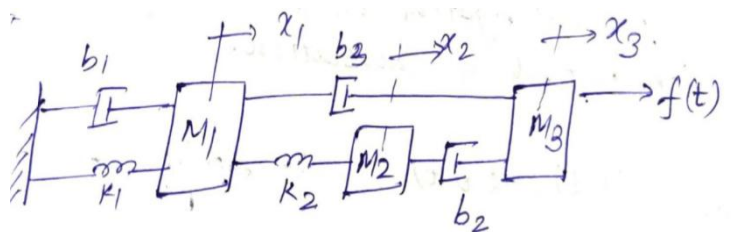


Fig. Q4
